

DL (14th Feb)

CNN → learn spatial hierarchies of features
↳ basic blocks → filters (kernels)

$$y(m,n) = u(m,n) * h(m,n) = \sum_i \sum_j u(i,j) h(m-i, n-j)$$

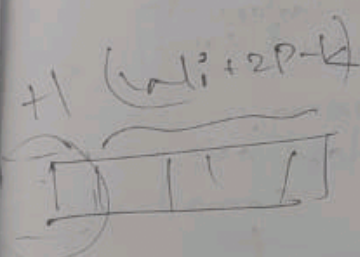
→ (HxW) input $M \times N$ filter → valid output: $(H-M+1) \times (W-N+1)$
✓ full output: $(H+M-1) \times (W+N-1)$

but here it doesn't $\leq 180^\circ$ flip
make diff.

to prop. of conv. operations:
→ commutative $u * h = h * u$
→ associative $h_1 * (h_2 * u) = (h_1 * h_2) * u$
→ distributive $(u_1 + u_2) * h = u_1 * h + u_2 * h$
Note: edge padding → to compute response of edges.

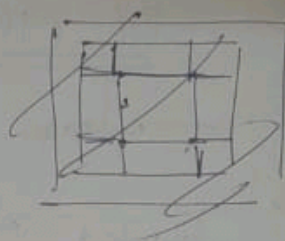
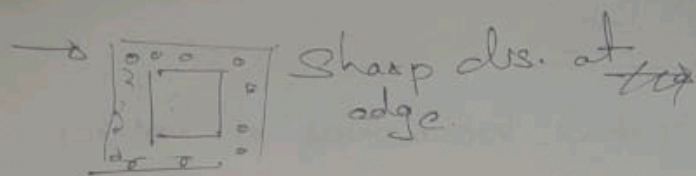
(i) $w_o = \frac{w_i + 2P - K}{S} + 1$

w_o → size of output
 w_i → size of input
 P → padding size
 K → kernel/filter size
 S → stride (point of shift)
 $+1$ → no. of steps by which the filter slide over input.



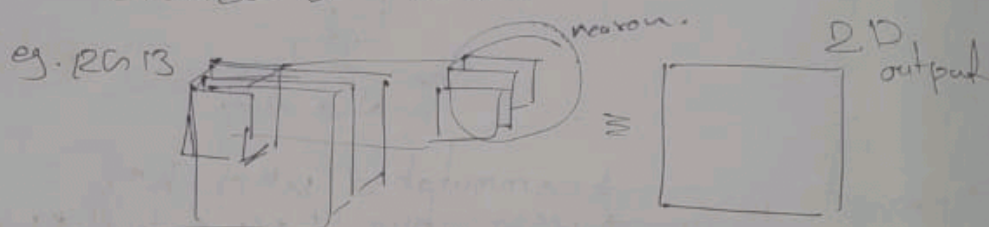
(ii) $w_i + 2P - K + 1 = w_i \Rightarrow P = \frac{K-1}{2}$ for input = output

(iv) ignore padding → edge not imp as info is centralised
(v) zero constant padding, explicit/edge padding, reflective padding, Circular padding.



→ Circular padding →

Note: No. of channels of input must match no. of channels of convolutional filter.



2D convolution, i.e.

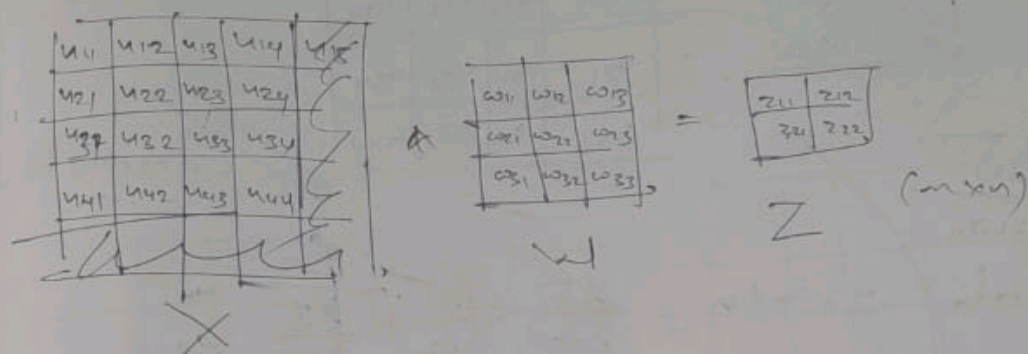
$$w \otimes u + w \otimes v + w \otimes s$$

use of convolutional filters.

- spatial invariance.
- preservation of spatial structure (i.e. relative position of feature in input & output same)
- parameter sharing (i.e. sets of filters are same).
- sparsity of connections.

* output has multiple channels when multiple convolutional filters are used

Back propagation through 'Convolutional Filters'



$$\rightarrow \frac{\partial J}{\partial w_{ij}} = \frac{\partial J}{\partial z} \frac{\partial z}{\partial w_{ij}}$$

$$\frac{\partial J}{\partial w_{ij}} = \sum_m \sum_n \frac{\partial J}{\partial z_{mn}} \frac{\partial z_{mn}}{\partial w_{ij}} \quad (\text{because } z_{mn} \text{ depends on every } w_{ij}).$$

$$z_{11} = u_{11}w_{11} + u_{12}w_{12} + u_{13}w_{13} + u_{21}w_{21} + u_{22}w_{22} + u_{23}w_{23} + u_{31}w_{31} + u_{32}w_{32} + u_{33}w_{33}$$

$$z_{12} = u_{12}w_{11} + u_{13}w_{12} + u_{14}w_{13} + u_{22}w_{21} + u_{23}w_{22} + u_{24}w_{23} + u_{32}w_{31} + u_{33}w_{32} + u_{34}w_{33}$$

$$z_{21} =$$

$$z_{22} =$$

$$\frac{\partial J}{\partial w_{11}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{11}}$$

$\frac{\partial J}{\partial z_{11}}$	$\frac{\partial J}{\partial z_{12}}$
$\frac{\partial J}{\partial z_{21}}$	$\frac{\partial J}{\partial z_{22}}$

u_{11}	u_{12}	u_{13}	u_{14}
u_{21}	u_{22}	u_{23}	u_{24}
u_{31}	u_{32}	u_{33}	u_{34}
u_{41}	u_{42}	u_{43}	u_{44}

convolution

$$\frac{\partial J}{\partial w_{11}}$$

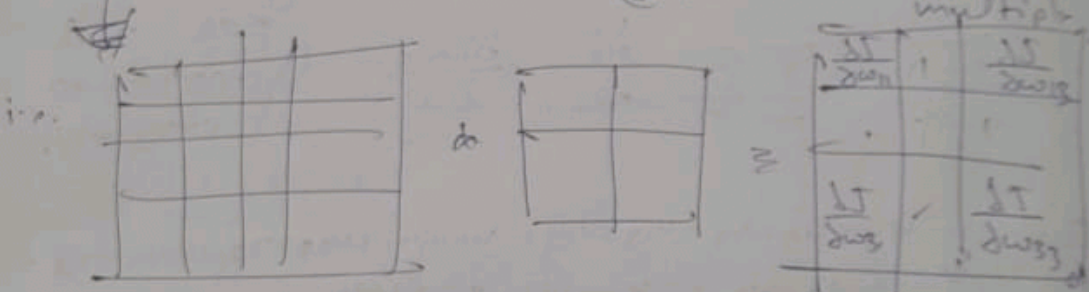
Note

$$z = w \cdot u$$

$$\frac{\partial J}{\partial w_{11}} = \left(\frac{\partial J}{\partial z} \right)^T \frac{\partial z}{\partial w_{11}}$$

$$= \left(\frac{\partial J}{\partial z} \right)^T u^T$$

multiply



$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial z} \cdot u$$

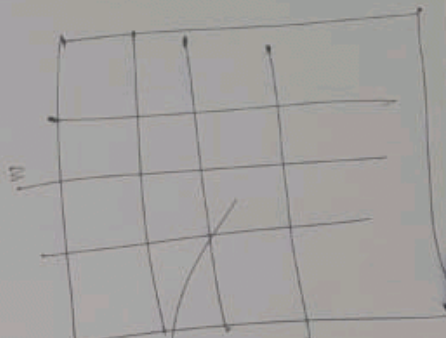
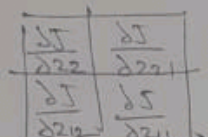
$$\frac{\partial J}{\partial x} = \frac{\partial J}{\partial z} \frac{\partial z}{\partial x}$$

$$\frac{\partial J}{\partial w_{ij}} = \sum_m \sum_n \frac{\partial J}{\partial z_{mn}} \frac{\partial z_{mn}}{\partial w_{ij}}$$

Not every z_{mn} will depend on every w_{ij}
 So term which don't depend on w_{ij} are 0.

$$\frac{\delta J}{\delta u_{12}} = \frac{\delta J}{\delta z_{11}} \omega_{12} + \frac{\delta J}{\delta z_{12}} \omega_{11} + \frac{\delta J}{\delta z_{21}} \frac{\delta z_{21}}{\delta u_{12}} + \frac{\delta J}{\delta z_{22}} \frac{\delta z_{22}}{\delta u_{12}}$$

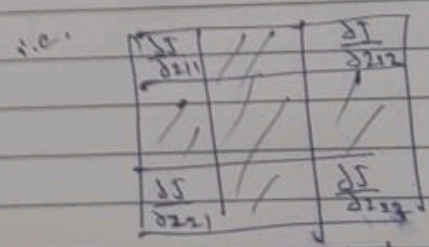
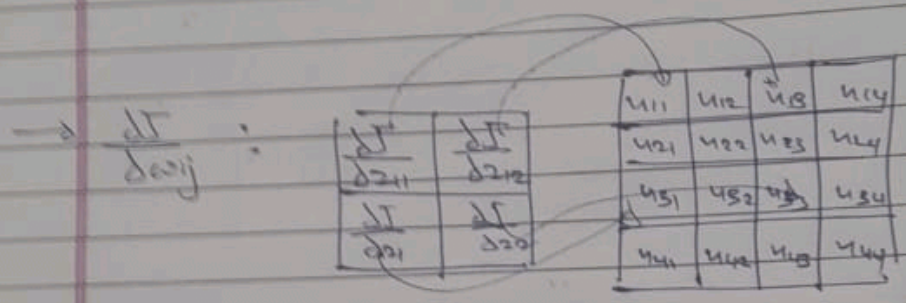
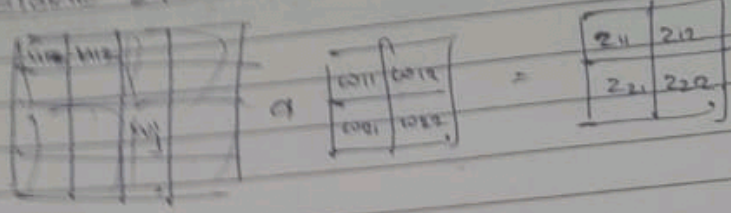
$$\frac{\delta J}{\delta u_{22}} = \frac{\delta J}{\delta z_{11}} \omega_{22} + \frac{\delta J}{\delta z_{12}} \omega_{21} + \frac{\delta J}{\delta z_{21}} \omega_{12} + \frac{\delta J}{\delta z_{22}} \omega_{11}$$



$$\frac{\delta J}{\delta}$$

$$\frac{\delta J}{\delta w_{ij}}$$

→ state-2:



diagonal convolution
(add zeros in kernel)
(adding gaps)

diagonal kernel

$$\frac{\partial J}{\partial w_{11}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{11}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{11}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{11}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{11}}$$

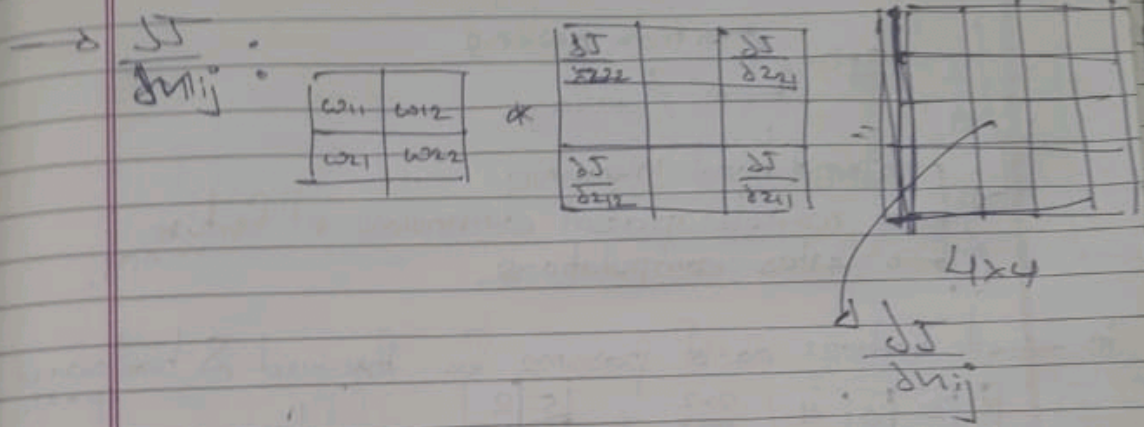
$$= \frac{\partial J}{\partial z_{11}} u_{11} + \frac{\partial J}{\partial z_{12}} u_{13} + \frac{\partial J}{\partial z_{21}} u_{31} + \frac{\partial J}{\partial z_{22}} u_{33}$$

Note: diagonal route.



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10. Just disable the upstream gradient



pooling → are applied to features, to reduce spatial dimension
layers:

types → max pooling
Average

adv. 2.9.
→ translational invariance
→ reduces spatial dimensions of feature maps.
→ faster computations.

ie. → it reduces no. of params for forward & backward pass.

1	3	0	4	2x2 max pooling	5	8
5	0	8	4		7	6
7	0	2	1			
0	1	0	6	2x2 avg pooling	2.25	3.5
					2	2.25

focus on imp details.

Note: translational invariance → in local neighbourhood
ie. exact pos of max doesn't matter.
orig val is imp

Max Pooling

				max elements.

2x2 max pooling

2 ₁₁	2 ₁₂
2 ₂₁	2 ₂₂

X → Z

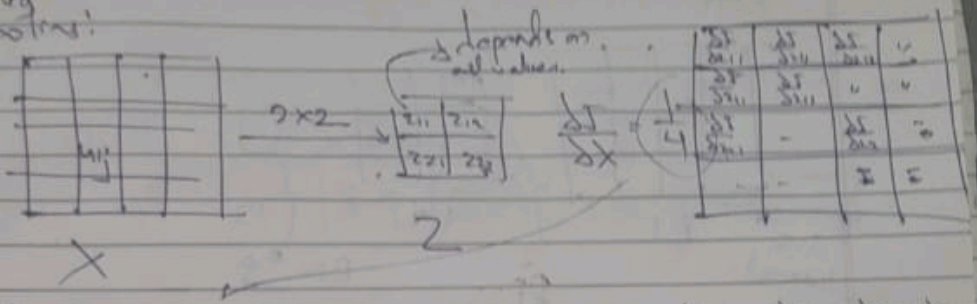
$\frac{\partial J}{\partial x} =$

0	0	0	$\frac{\partial J}{\partial z_{22}}$
0	$\frac{\partial J}{\partial z_{11}}$	0	0
$\frac{\partial J}{\partial z_{21}}$	0	0	0
0	0	$\frac{\partial J}{\partial z_{22}}$	0

Similar to max/min ~~other~~ operations
gradient sensitivity

$z_{11} = u_{12}$ $\frac{\partial L}{\partial u_{12}} = \frac{\partial L}{\partial z_{11}} \frac{\partial z_{11}}{\partial u_{12}} + \dots = \frac{\partial L}{\partial z_{11}}$

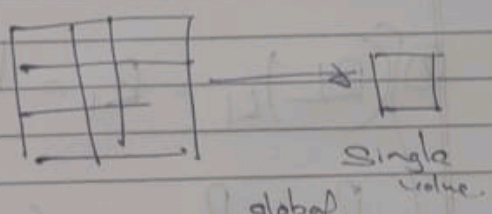
→ Avg pooling:



bc2, $z_{11} = \frac{1}{2} (u_{11} + u_{12} + u_{21} + u_{22}) \Rightarrow \frac{\partial z_{11}}{\partial u_{11}} = \frac{\partial z_{11}}{\partial u_{12}} = \frac{\partial z_{11}}{\partial u_{21}} = \frac{\partial z_{11}}{\partial u_{22}} = \frac{1}{4}$

$\frac{\partial L}{\partial u_{11}} = \frac{\partial L}{\partial z_{11}} \frac{\partial z_{11}}{\partial u_{11}} + \frac{\partial L}{\partial z_{12}} \frac{\partial z_{12}}{\partial u_{11}} + \dots$

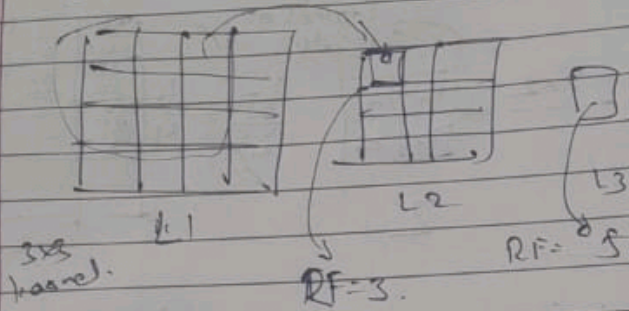
* GAP → global avg pooling.



Global: $(W \times H \times C) \xrightarrow[\text{p.}]{\text{global avg.}} 1 \times 1 \times C$

Handwritten notes: "channel", "global avg.", "single value."

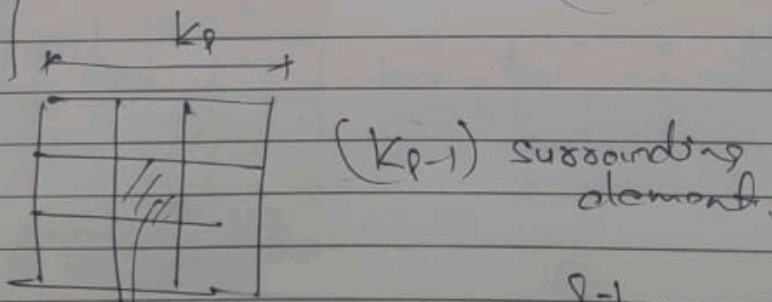
Receptive fields of ~~conv~~ layers is $(RF)^{CNN}$ area visible to this layer.



Note: deeper layers $\rightarrow$ effectively look at larger area of input, that shallower region.

① $(RL)_{L1} - R_1 = k_1 \rightarrow$ size of kernel of 1st layer

$R_k = R_{k-1} + (k_k - 1) \prod_{i=1}^{k-1} S_i \Rightarrow$ effective stride upto layer k



$R_{k-1} + (k_k - 1) \prod_{i=1}^{k-1} S_i$

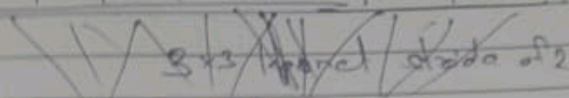
Note: $\{k_k \rightarrow$ is used to form R_k (Layer $k\}$



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eg.

1	2	3	4	5	6	7	8	9
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$k_1 = 3$
 $s_1 = 2$

1-3	3-5	5-7	7-9
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L1 $R_1 = 3$

$k_2 = 3$
 $s_2 = 1$

1-7	3-9	5-9
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L2 $R_2 = 7$

$$R_2 = R_1 + (3-1)(2 \times 1) = 3 + 4 = 7$$